Assessment_01_CO_02

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QUESTIONS

- **Q1** Check Whether a Number Is an Abundant Number — 5 marks
- **Q2** Check Whether a Number Is a Pronic Number — 5 marks
- **Q3** Check Whether a Number Is a Münchhausen Number — 5 marks
- **Q4** Check Whether a Number Is a Powerful Number — 5 marks

4/4 answered

Q1 Check Whether a Number Is an Abundant Number

5 marks

Topics: Classes, objects, access specifiers, member functions, loops, divisors, arithmetic calculations.

A positive integer is called an Abundant number if the sum of its proper divisors is greater than the number itself.

Proper divisors are the positive divisors excluding the number itself.

For example, the proper divisors of 12 are 1, 2, 3, 4, and 6.

Their sum is:

\[
1+2+3+4+6=16
\]

Since \(16>12\), the number 12 is an Abundant number.

Write a C++ program using a class Abundant to read a number and check whether it is an Abundant number.

Test case: 12->Abundant number
15 → Not an Abundant number.

Your Code

```cpp
#include <iostream>
using namespace std;

class Abundant
{
public:
    int n, sum = 0;

    void get()
    {
        cin >> n;
    }

    void check()
    {
        for (int i = 1; i < n; i++)
        {
```

Input

```
15
```

Output

```
Not an Abundant number
```

Visible Test Cases

Test Case 1

Your Code

```cpp
#include <iostream>
using namespace std;

class Abundant
{
public:
    int n, sum = 0;

    void get()
    {
        cin >> n;
    }

    void check()
    {
        for (int i = 1; i < n; i++)
        {
            if (n % i == 0)
            {
                sum = sum + i;
            }
        }

        if (sum > n)
            cout << "Abundant number";
        else
            cout << "Not an Abundant number";
    }
};

int main()
{
    Abundant a;
    a.get();
    a.check();

    return 0;
}
```

Submitted

Input

```
15
```

Output

```
Not an Abundant number
```

Visible Test Cases

Test Case 1

Test Case 2

5.00 / 5 marks

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QUESTIONS

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- Q4 ✓ Check Whether a Number Is a Powerful Number — 5 marks

4/4 answered

Q2 Check Whether a Number Is a Pronic Number

5 marks

Topics: Classes, objects, private data members, public member functions, loops, multiplication, logical conditions.

A number is called a Pronic number if it can be expressed as the product of two consecutive integers.

For example:
- \(12=3\times4\) — Pronic number.
- \(20=4\times5\) — Pronic number.
- 15 cannot be expressed as the product of two consecutive non-negative integers, so it is not a Pronic number.

Write a C++ program using a class Pronic to read a number and check whether it is a Pronic number.

Test case: 20->Pronic number
15 → Not a Pronic number.

Your Code

```cpp
#include <iostream>

using namespace std;

class Pronic {
private:
    int number;

public:
    void readNumber() {
        cin >> number;
    }

    void checkPronic() {
        if (number < 0) {
            cout << "Not a Pronic number" << endl;
            return;
        }

        bool isPronic = false;
        int i = 0;

        while (i * (i + 1) <= number) {
```

Input

```
20
```

Output

```
Pronic number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓

5.00 / 5 marks

4/4 answered

```cpp
using namespace std;

class Pronic {
private:
    int number;

public:
    void readNumber() {
        cin >> number;
    }

    void checkPronic() {
        if (number < 0) {
            cout << "Not a Pronic number" << endl;
            return;
        }

        bool isPronic = false;
        int i = 0;

        while (i * (i + 1) <= number) {
            if (i * (i + 1) == number) {
                isPronic = true;
                break;
            }
            i++;
        }

        if (isPronic) {
            cout << "Pronic number" << endl;
        } else {
            cout << "Not a Pronic number" << endl;
        }
    }
};

int main() {
    Pronic obj;
    obj.readNumber();
    obj.checkPronic();
    return 0;
}
```

Submitted

Output

```
Pronic number
```

Visible Test Cases

Test Case 1

Test Case 2

5.00 / 5 marks

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QUESTIONS

- Q1 ✓ Check Whether a Number Is an Abundant Number — 5 marks
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- Q4 ✓ Check Whether a Number Is a Powerful Number — 5 marks

4/4 answered

Q3 Check Whether a Number Is a Münchhausen Number

5 marks

Topics: Classes, objects, access specifiers, member functions, digit extraction, loops, and exponentiation.

A Münchhausen number is a number that is equal to the sum of its digits, with each digit raised to the power of itself. For this program, consider \(0^0=0\).

For example, the number 1 is a Münchhausen number because \(1^1=1\).

Write a C++ program using a class Munchhausen to read a number and check whether it is a Münchhausen number.

Test case: 1-> Munchhausen number
10 → Not a Munchhausen number.

For 10, the sum is \(1^1+0^0=1\), which is not equal to 10.

Your Code

```cpp
#include <iostream>
#include <cmath>

using namespace std;

class Munchhausen {
private:
    int number;

    int powerSelf(int digit) {
        if (digit == 0) {
            return 0;
        }
        return pow(digit, digit);
    }

public:
    void readNumber() {
        cin >> number;
    }

    void checkMunchhausen() {
        if (number == 0) {
```

Input

```
1
```

Output

```
Munchhausen number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓

5.00 / 5 marks

```
        }
        return pow(digit, digit);
    }

public:
    void readNumber() {
        cin >> number;
    }

    void checkMunchhausen() {
        if (number == 0) {
            cout << "Munchhausen number" << endl;
            return;
        }

        int temp = number;
        int sum = 0;

        while (temp > 0) {
            int digit = temp % 10;
            sum += powerSelf(digit);
            temp /= 10;
        }

        if (sum == number) {
            cout << "Munchhausen number" << endl;
        } else {
            cout << "Not a Munchhausen number" << endl;
        }
    }
};

int main() {
    Munchhausen obj;
    obj.readNumber();
    obj.checkMunchhausen();
    return 0;
}
```

Visible Test Cases

Test Case 1

Test Case 2

5.00 / 5 marks

Submitted

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4/4 answered

Q4 Check Whether a Number Is a Powerful Number — 5 marks

Topics: Classes, objects, private and public access specifiers, member functions, loops, prime factorization.

A positive integer is called a Powerful number if every prime factor of the number occurs at least twice in its prime factorization.

For example:
- \(36=2^2\times3^2\) — Powerful number.
- \(12=2^2\times3\) — Not a Powerful number because the prime factor 3 occurs only once.

Write a C++ program using a class Powerful to read a number and determine whether it is a Powerful number. Treat 1 as a Powerful number because it has no prime factors violating the condition.

Test case: 36->Powerful number
12 → Not a Powerful number

Your Code

```cpp
#include <iostream>

using namespace std;

class Powerful {
private:
    int number;

public:
    void readNumber() {
        cin >> number;
    }

    void checkPowerful() {
        if (number <= 0) {
            cout << "Not a Powerful number" << endl;
            return;
        }

        if (number == 1) {
            cout << "Powerful number" << endl;
            return;
        }
```

Input

```
12
```

Output

```
Not a Powerful number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓

5.00 / 5 marks

```
        if (number <= 0) {
            cout << "Not a Powerful number" << endl;
            return;
        }

        if (number == 1) {
            cout << "Powerful number" << endl;
            return;
        }

        int temp = number;
        bool isPowerful = true;

        for (int i = 2; i * i <= temp; i++) {
            if (temp % i == 0) {
                int count = 0;
                while (temp % i == 0) {
                    count++;
                    temp /= i;
                }
                if (count < 2) {
                    isPowerful = false;
                    break;
                }
            }
        }

        if (temp > 1) {
            isPowerful = false;
        }

        if (isPowerful) {
            cout << "Powerful number" << endl;
        } else {
            cout << "Not a Powerful number" << endl;
        }
    }
};

int main() {
    Powerful obj;
    obj.readNumber();
    obj.checkPowerful();
    return 0;
}
```

Test Case 1

Test Case 2

5.00 / 5 marks